Using Transit Network Connectivity, Geography and Demographics to Predict Station-Level Daily Ridership in Tokyo

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Abstract

The city of Tokyo is home to one of the densest, busiest, and most complex rail networks in the world. It plays host to a variety of operators and agencies that transport millions of residents and tourists through the city. Understanding what drives station-level demand has implications for urban planning and transit investment. In this project, we investigate the connectivity characteristics of the city's transit network, along with their demographic and geographic context, to determine whether they can be used to predict daily use at the station-level.

We first preprocess and synthesize data from a variety of sources, including GTFS schedules, ridership statistics from MLIT, and e-Stat census tracts. Using this data, we then create a feature matrix of 474 individual stations across the citys 23 special wards. In particular, the GTFS data is used to derive route count, service frequency, and transfer distance to the Yamanote Line, which is used as a reference point due to it traversing through many of the citys busiest stations and commercial hubs. All models were evaluated under spatial cross-validation to avoid information leakage from spatial autocorrelation. After comparing model performance across a mean-predictor baseline, Ridge regression, Random Forest, and Gradient Boosting, the Random Forest was determined to have the strongest predictive performance. Analysis of feature importance reveals that daily stop events, mean route centrality, and distance to a station visited by the Yamanote Line constitute the primary drivers behind the models explanatory power compared to residential population. This is likely due to the busiest stations being in busy commercial areas.

Finally, the question of transit equity is reframed to be a matter of prediction error. Namely, stations where the model frequently underestimates or overestimates ridership deserve further scrutiny.

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1 Introduction

Understanding what drives demand at individual stations is instrumental in determining how to allocate resources effectively. This can determine if service frequency must be increased, if infrastructure has to be improved, or if existing capacity is underutilized. This matter is non-trivial in Tokyos vast and complex rail network, as ridership varies massively across different stations in varying levels of population density.

Many naive approaches to this question tend to be flawed. Such approaches often focus on population density around a station, however residential population alone cannot explain ridership demand, as the majority of the busiest stations lie in commercial districts rather than residential districts. It also cannot be reasonable to conclude that stations that lie on major lines are busier, since it does not explain how much of this variation is influenced by network position, along with the surrounding geography and demographics. Determining what drives daily ridership requires a multi-source engineering approach paired with a rigorous evaluation framework that takes into account several variables rather than focusing on a single one.

Using data gathered from GTFS schedule data for the Kanto region, ridership statistics for the fiscal year of 2019 as reported by MLIT, and census-tract population from the 2020 National Census, we investigate whether connectivity features, along with the geographic and demographic context can predict station-level daily ridership in the city. We derive connectivity features including route count, service frequency, amount of transfers to the Yamanote Line, and centrality on a route-level graph, and the total population within 800 meters of each station. We then apply spatial cross-validation to prevent leaking spatial information. Under this spatial cross-validation protocol, we compare performance between a mean-predictor baseline, Ridge regression, Random Forest and Gradient Boosting. The prediction errors are then used to revisit the question of which areas of Tokyo may be under or over-served by the rail network.

2 Related Work

2.1 ML Approaches in transit ridership

Public transport has evolved to be a popular testing ground for machine learning applications. A 2024 paper (1) implemented a combination of Backward Elimination, XGBoost, and SHAP Analysis to investigate the influence of the built environment surrounding subway stations on passenger flow, and whether time of day affects the degree of influence. (9) expanded on the problem, analyzing the relationship over multiple years using a combination of LightGBM and SHAP analysis. It was discovered here that the relationship is a non-linear one. (5) attempted to investigate whether image representations of land use and modelling trips at the origin-destination level can improve urban rail ridership prediction. The paper uses a version of a CNN designed specifically to analyze origin-destination pairs, and it was demonstrated that this CNN outperformed baseline models due to lower prediction error and a better capture of spatial complexity.

2.2 Network Theoretic analysis of transit systems

The complexity of Tokyo's railway network makes it a popular case study for research in public transport design, which can later be used to guide future infrastructure projects. (6) analyzed the design of Tokyos railway stations from looking at their node value, which is the strength of the station as a part of the rail network, and place value, the strength of the surrounding area. The paper then examines the relation between these two values and actual ridership. For each station in their dataset, node and place indexes were calculated using information entropy weighting. The study later concludes that while both values increase ridership, node value has five times more influence over ridership than place value. A 2023 study (7) focused on the relationship of demographics and public transport. Specifically, it aimed to investigate whether younger and older people receive equal access to high-quality transit-centric environments. Stations that are visited by JR Easts busiest lines, the Yamanote and the Chuo, were used as case studies. The data was acquired through mobile phone data and through methods involving the Gini coefficient and correlation analysis, it was discovered that better transit-oriented development environments tend to align with younger users than older users. In (8), a study analyzed the effect on land use around JR commuter stations on ridership. In particular, the Passenger Catchment Area of each station was analyzed. The study uncovered that the land use within 500 meters of the station best explains ridership, as 400-800 meters is considered as the standard walking range to transit.

3 Methodology

3.1 Data Preprocessing

As mentioned earlier, this project utilizes data from a raw set of GTFS data, MLIT ridership statistics from the fiscal year of 2019, and e-Stat census data from 2020. The GTFS feed in particular was sourced from the TokyoGTFS project (GitHub), which covers the city and the surrounding Kanto region. The data produced by this repository is derived from data published by the Open Data for Public Transportation initiative. For this project, we were interested in the `stops.txt`, `stop_times.txt`, `trips.txt`, and `routes.txt` files. These files detail what stops, trips, and routes exist in the city, as well as the train schedules pertaining to them.

As we are primarily interested in the area surrounding Tokyos 23 wards, the initial preprocessing step filtered the dataset to stations within those wards, and retained only the relevant data accordingly. In `stops.txt`, each line that stops at a given station has a separate record for said station. For example, if the Yamanote Line stops at Tokyo Station, then the dataset contains both a record for the station itself, `Tokyo`, and the line-specific record, `Yamanote.Tokyo`. Since this structure can create multiple entries for a single station, we created a unified station key for each by extracting the English portion of the bilingual `stop_name` field (where the format is <Kanji Name><English Name> and collapsing records under said key. Each station was also converted to a point geometry and spatially joined to Tokyos 23 special-ward polygons using an `intersects` predicate.

The ridership records were sourced from MLITs National Land Numerical Information. As there are separate ridership reports from each operator, we aggregated the MLIT records by their Japanese station name and summed the figures across all operators. These names were matched to GTFS stations using the kanji component of `stop_name`. Stations that were not linked were excluded from the final dataset.

The raw file from the 2020 census-tract from e-Stat required Shift-JIS decoding and a removal of a metadata row embedded after the header, and filtering to the finest aggregation level (chome(district) level) to avoid double-counting parent rows at ward and oaza(subdivision) levels. The total population counts within the 800 m Euclidean radius around each station were computed.

3.2 Feature Preprocessing

3.2.1 Network connectivity features

We constructed a route-level graph $G = (V, E)$ where each node is a GTFS route. An undirected edge connects two routes whenever they share at least one physical station. This representation was chosen due to it encoding the transfer structure of the network: two routes that meet at a hub are adjacent irrespective of how many intermediate stops each route serves. Below are the following feature families derived from this group.

Yamanote transfer distance: A Breadth-first search (BFS) was run from the Yamanote Line node. For each station, the minimum shortest-path length among all routes serving it was recorded and yields `min_transfers_yamanote`. This number counts the least amount of route-to-route transfers required to reach the Yamanote Line. The Yamanote Line was chosen as a reference point for this study as it traverses Tokyos busiest stations, and in this light, serves as a boundary for Tokyos most populated area.

Graph Centrality: For G, we computed degree and betweenness centralities. For each station, we then aggregated across its serving routes to produce the relevant columns. The degree centrality describes how many transfer opportunities a given route offers, while betweenness centrality describes routes that serve as bridges between otherwise poorly connected parts of the network.

Route count: This is the number of routes serving each station.

3.2.2 Service intensity

We calculated the total amount of scheduled stop events per station across all routes and trips in the GTFS schedule. This is detailed under `daily_stop_events`.

3.2.3 Geographic and demographic features

The vectorized haversine formula was used to measure the great-circle distance from each station to the nearest station serving the Yamanote. The distances are under the column `km_to_yamanote`. This adds to the transfer-count feature by describing the spatial proximity independently of the network topology, as two stations may be considered a transfer from the Yamanote but differ substantially in physical distance. `ward_jp`, the categorical ward assignment, was one-hot encoded. `Pop_800m_catchment` was computed from chome-level census data.

3.2.4 Target Variable

The busiest station, Shinjuku, exceeds the median by approximately two orders of magnitude. A $\ln(1+y)$ transformation was applied for variance stability and to produce an approximately normal target distribution. This improves performance for linear models and simplifies residual interpretation. Below are the evaluation metrics reported on both the log and original scales.

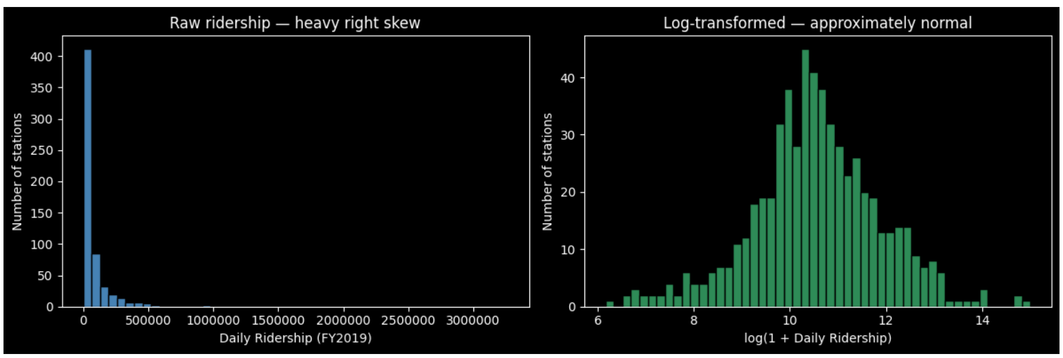

Figure 1: Tokyo Station during peak hours.

3.3 Models

For this project, we compared the performance of four models of increasing complexity in determining the relationship between features and ridership. `DummyRegressor`, our mean baseline that always predicts the training-set mean, was used as a benchmark for all other models. The Ridge Regression (where α = 1.0) aims to determine whether a linear mapping from features to log-ridership is sufficient. L2 Regularization was used here because many connectivity features exhibit pairwise correlations above 0.6, and Ridge prevents coefficient inflation under this collinearity. The Random Forest, which uses 300 trees with minimum leaf size 3 for this case, aims to discover whether non-linear interactions, such as the potential effect of service frequency depending on distance to the Yamanote Line, can yield further gains. Gradient Boosting, with 300 estimators, depth of 4, and learning rate of .05, applies sequential residual correction and normally represents the strongest conventional learner on tabular data.

The models were wrapped in a unified preprocessing pipeline, where numeric features were standardized using `StandardScaler`, and the categorical ward variable was one-hot encoded. The pipeline was applied identically within each cross-validation fold to prevent information leakage from the test partition into the scaling statistics.

For evaluation, R^2 on the log-transformed target served as the primary metric of explained variance. The Mean absolute error (MAE) on the log scale provided a scale-interpretable average miss. The Root mean squared error (RMSE), which was computed after back-transforming predictions to the original passengers-per-day scale via `expm1`, described error magnitude in operationally meaningful units. These metrics were reported as mean + standard deviation across all eight spatial folds.

The best-performing model was supplemented by two interpretability layers. The first layer added were SHAP values, which were computed to decompose each prediction into per-feature contributions in order to uncover which features most consistently drive the models output across the dataset. The other were the out-of-fold residuals, which were defined as actual minus predicted log-ridership, both in terms of geography and ward.

4 Experiments & Results

4.1 Model Comparison

Table 1: Model Performance Comparison

Model	R^2	MAE(log)	RMSE(orig)
Mean (baseline)	-0.088 ± 0.077	1.035	213,533
Ridge	0.527 ± 0.072	0.677	1,535,286
Random Forest	0.679 ± 0.076	0.558	123,809
Gradient Boost	0.671 ± 0.107	0.561	123,161

Table 1 above summarizes the cross-validated performance of the four models. The mean and standard deviation are computed across the eight spatial folds. Based on the comparison, the mean baseline produces a negative R^2, confirming a substantial variance of ridership across stations, and that a constant prediction performs badly under spatial cross-validation. The Ridge regression achieves a R^2 of .527 on the log scale, showing that even a linear combination of the features could capture a meaningful level of variance. While this is an improvement over the baseline, the Ridge regressions back-transformed RMSE is an order of magnitude much larger than the other models, at around 1.54 million passengers a day. This is due to the limitation of Ridge being unable to model non-linear relationships, which tends to produce extreme mispredictions for a number of very-high-ridership stations. The Random Forest and Gradient Boosting have similar performance on log-scale MAE, but Random Forest has a slightly higher R^2, and exhibits lower fold-to-fold variance, suggesting that it is more stable in generalizing across spatial partitions. Thus, the Random Forest was selected as the best-performing model.

4.2 Random Forest

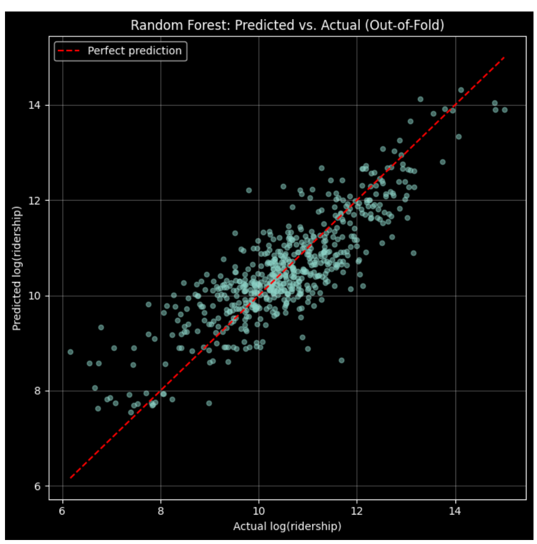

Figure 2: Random Forest's out-of-fold predictions.

In order to examine the predictions of the Random Forest in more detail, we collected out-of-fold predictions across all eight folds, yielding one predicted value per station. The overall out-of-fold R^2

computed from these predictions is 0.708, which is slightly higher than the mean fold-level R^2 of .679.

The predicted-versus-actual scatter plot above shows the data clustering along the diagonal across the majority of the ridership range from small stations near log-ridership 6 to mid-sized stations near 10. In terms of dispersion, the model has a tendency to over-predict for the lowest-ridership stations, which are below log-ridership 7, and under-predict for the highest ridership stations, which are above log-ridership 12. This pattern is consistent with the general difficulty of predicting tail behavior in heavily skewed distributions even after log-transformation.

4.3 Feature Importance

As mentioned in section 3.3, SHAP and feature importance from the tree ensemble were used to assess which features drive the predictions of the Random Forest.

From the Random Forests built-in importances, it was revealed that the service frequency proxy, `daily_stop_events`, accounts for nearly 63% of total importance, making it the most dominant predictor. This was followed by the mean degree centrality of routes serving a station, `mean_route_degree_cent`, and geographic distance to the nearest station served by the Yamanote, `km_to_yamanote`. In particular, the one hot encoded ward indicators collectively contribute very little, which suggests that the model is learning generalizable patterns based on continuous features rather than memorizing ward-level averages.

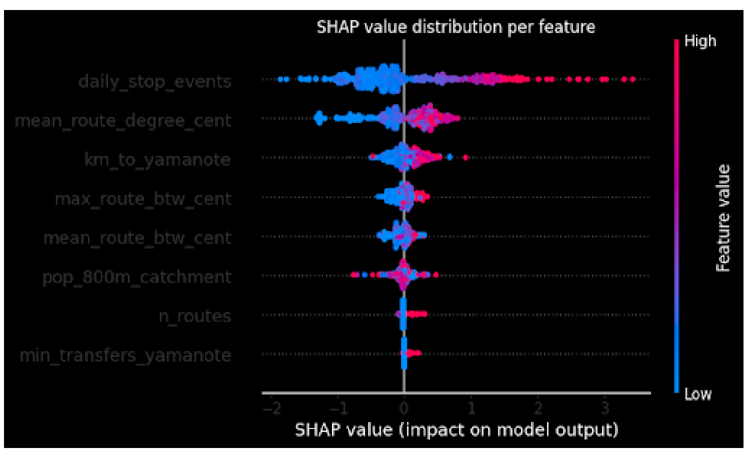

Figure 3: Random Forest's out-of-fold predictions.

The SHAP beeswarm plot above shows that high values of `daily_stop_events` and `mean_route_degree_cent` push predictions upward, as expected. Being further from a station served by the Yamanote line, as represented by a greater `km_to_yamanote`, consistently pushes predictions downward. The modest ranking of the demographic feature `pop_800m_catchment` indicates that while population density has a positive relation with ridership, it contributes less predictive power than network-level connectivity and service frequency. This assessment is consistent with the observation that many of Tokyos highest-ridership stations are located in busy commercial districts with a low nighttime residential population, and that ridership is driven more by the stations role in the network than the surrounding population.

4.4 Residual Analysis

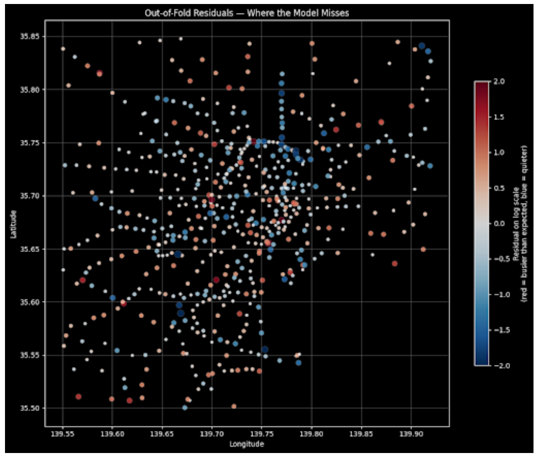

Figure 4: Random Forest's out-of-fold residuals showing the model's misses.

To analyze these results to the secondary research question - which areas of Tokyo may be over or underserved by the rail network - we examine the distribution of out-of-fold residuals, defined as actual log-ridership minus the predicted log-ridership. A positive residual suggests that a station is busier than what the model predicted. Conversely, a negative residual suggests that a station is quieter than expected.

The stations with the highest positive residuals include the Tokyo Monorails Hamamatsucho station, Kyobashi station, and Sakaecho station. These stations tend to function as transfer hubs for specific corridors such as the Tokyo Monorail to Haneda airport or serve major commercial destinations whose ridership is driven by factors beyond our feature set such as tourism or business. Several of the most over-predicted stations are on the Toden Arakawa Line (Sakura Tram), particularly stations in Arakawa ward. These include Arakawa-nanachome, Arakawa-itchumae and Arakawa-nichome. These stations have high scheduled stop-event counts relative to the actual amount of passengers served due to the streetcar having a high frequency but short, localized trips. Since the model treats stop-event frequency as a strong ridership signal, it overestimates demand at these stations.

On the ward level, Arakawa ward has the most negative mean residual, followed by the Taito and Adachi wards. These are located in Tokyos northeast regions, where many stations are quieter than their network features would suggest. On the opposite end, the Edogawa and Chuo wards have the most positive mean residuals, indicating that their stations are generally busier than predicted. Stations beyond the 23 special wards also show a positive mean residual of +.251. This is likely due to commuter rail termini drawing ridership from a wider area than the 800-meter walking catchment captures.

These patterns also offer a data-driven lens on transit equity. Wards with persistently negative residuals may represent areas where the transit network provides more capacity than the actual demand warrants, or where competing modes of transit such as walking, cycling or buses, reduce reliance on rail. Conversely, wards with positive residuals are potential candidates for service expansion.

5 Conclusion

This project aims to determine whether network connectivity, geography and demographics can predict station-level daily ridership in Tokyo, and eventually, the results were used to determine which areas of Tokyo were underserved or overserved by public transit.

From this study, the most striking finding was the degree to which network-level features dominate the prediction of station ridership. The Random Forests impurity-based importances and the SHAP analysis revealed that daily stop events and mean route degree centrality both account for the vast majority of explanatory power, while the demographic feature contributes relatively little. It is natural for one to expect that the level of population living near a station would be a strong predictor of its ridership. However, Tokyos highest-ridership stations are mainly located in commercial and employment districts with a low nighttime residential population. Shinjuku is a notable example. It is the busiest station in Tokyo and in the world, yet has a modest immediate residential catchment relative to its ridership.

This finding also carries a practical implication for transit planning. If ridership is primarily shaped by the connectivity of a station rather than the level of the nearby population, then design and investment decisions focused narrowly on population projects may underestimate demand at well-connected interchange stations and overestimate it at stations in dense residential areas that have a weak network position.

In addressing the question of which areas of Tokyo may be overserved or underserved by the railway network by way of the prediction errors, we have uncovered that stations where actual ridership consistently exceeds the models prediction are potentially underserved, while stations with a lower ridership than predicted are potentially overserved. The former type does not necessarily indicate the stations service is insufficient, but that the station attracts more riders through other means beyond the features we have analyzed, possibly from tourism, the presence of major commercial destinations, or the stations role as an interchange point. Conversely, the latter does not imply that stations such as those along the Toden Arakawa streetcar line are oversupplied, but rather the nature of the trip patterns along it - short, frequent, localized trips with low boarding volumes per trip - inflate the amount of stop events. The model operates under the assumption that a high stop-event frequency generally indicates high ridership, so it has the potential to overestimate demand at such stations due to it not being able to discern the ridership-generating potential of a Shinkansen stop event and that of a streetcar.

6 Discussion

The study has several limitations that should be acknowledged. For example, the target variable, the FY2019 daily ridership, represents a single pre-COVID snapshot. This year was chosen primarily due to it describing normal usage patterns not disrupted by the behavioral shifts caused by the pandemic, a single year cannot capture temporal trends such as the growth in ridership at stations near new residential developments or line closures. Also, the census and the ridership data are both from the pre-pandemic era, whereas the GTFS feed acquired is as of March 2026.

The connectivity was measured in terms of transfer counts and graph-theoretic centrality but not in terms of actual walk time. This was due to the GTFS data not having any data for footpaths nor transfer times. Platform layouts, headways and walking distances between platforms in a station have a drastic effect on travel time. In this light, a means of extending this project would be to add a travel-time-based accessibility measure derived from GTFS routing, should such data be available in the future.

There is also a limitation concerning the dominance of `daily_stop_events` in the feature importance rankings. Transit operators construct schedules based on demand, meaning high-ridership stations receive more service due to the fact that they have high ridership. The relation between stop events and ridership describes a feedback loop rather than a causal relationship. This does not invalidate the predictive model shown here, as stop-event frequency is a legitimate predictor, this fact complicates any causal interpretation of the features importance.

The study could also be extended to strengthen and generalize the findings of this study beyond the means suggested earlier. Expanding the analysis to include multiple years of ridership data could unlock the potential to study temporal trends and how demand grows and declines in each station over time. A panel-data approach could also be used to disentangle the endogeneity issue between service frequency and ridership. The study could also be extended to examine other metropolitan areas in Japan, such as Osaka, Nagoya, and Fukuoka to investigate whether the same pattern exists in those cities.

7 References

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Github Link: https://github.com/iidxDE/TokyoAccesibility